

Rain	Hagrid	Dumbl edore	$R \rightarrow H$	$H \vee D$	$\neg(H \wedge D)$	Knowle dge base
F	F	F	T	T	F	F
F	F	T	T	T	T	F
F	T	F	F	T	T	F
F	T	T	F	F	T	F
T	F	F	T	T	F	F
T	F	T	T	T	T	T
T	T	F	T	T	T	F
T	T	T	T	F	T	F

Task 1

$\neg \text{rain} \rightarrow \text{hagrid}$
 $\text{hagrid} \vee \text{dumbledore}$
 $\neg (\text{hagrid} \wedge \text{dumbledore})$
 dumbledore

Because the rest is false to it becomes true

since is Mustard is false and Plum is false then Scarlet is true
since is Kitchen is false and Ballroom is false then Library is true
since is Revolver is false and Wrench is false then Knife is true

Task 2

predict: false cause even when it rains harry will go to hagrid

run the code: false

why: without dumbledore we cant prove that is not raining so it prints false

who can be still placed in the house: Pomona and Gilderoy

gilderoy stays in ravenclaw, but minerva and pomona could be in either griffendor or hufflepuff

how to make it unique: not(pomonaGryffindor)

so pomona cannot join gryffindor which makes her a hufflepuff, while minerva is in gryffindor therefore now we only have one solution

Task 3

```
from logic import *

Aknight = Symbol("A is knight is: ")
Aknave = Symbol("A is knave is: ")

Bknight = Symbol("B is knight is: ")
Bknave = Symbol("B is knave is: ")

a_rules = And(
    Or(Aknight, Aknave),
    Not(And(Aknight, Aknave ))
)

b_rules = And(
    Or(Bknight, Bknave),
    Not(And(Bknight, Bknave ))
)

# puzzle 0
statement0 = And(Aknight,Aknave)

knowledge0 = And(
    a_rules,
    Implication(Aknight, statement0),
    Implication(Aknave, Not(statement0))
)

# puzzle 1
statement1 = And(Aknave, Bknave)

knowledge1 = And(
    a_rules,
    b_rules,
    Implication(Aknight, statement1),
    Implication(Aknave, Not(statement1))
)
```

```
87
88 # puzzle 2
89 same_kind = Or(
90     And(Aknight, Bknight),
91     And(Aknave, Bknave)
92 )
93 different_kind = Or(
94     And(Aknight, Bknave),
95     And(Aknave, Bknight)
96 )
97
98 knowledge2 = And(
99     a_rules,
100    b_rules,
101    Implication(Aknight, same_kind),
102    Implication(Aknave, Not(same_kind)),
103    Implication(Bknight, different_kind),
104    Implication(Bknave, Not(different_kind))
105 )
106
107
108 def show_result(title, knowledge, symbols):
109     print(title)
110     for symbol in symbols:
111         if model_check(knowledge, symbol):
112             print(" ", symbol)
113     print()
114
115 show_result("Puzzle 0", knowledge0, [Aknight, Aknave])
116 show_result("Puzzle 1", knowledge1,
117             [Aknight, Aknave, Bknight, Bknave])
118 show_result("Puzzle 2", knowledge2,
119             [Aknight, Aknave, Bknight, Bknave])
120
121
```

```
ghaliabadokhon@Ghalias-MacBook-Air lab3 % python3 knights.
Puzzle 0
  A is knave is:

Puzzle 1
  A is knave is:
  B is knight is:

Puzzle 2
  A is knave is:
  B is knight is:
```